

DEHAZE: AI-POWERED IMAGE DEHAZING USING DEEP LEARNING

Under the guidance of Prof. Suprava Patnaik
School of Electronics Engineering, KIIT (Deemed to be University), Bhubaneswar

Arnav Singhal (2330293), Abhishek Kumar singh (2330207), Bhavya Suman (2330298),
Anushka Sharma (2330291)

Keywords

MAXIM-S2, Image Dehazing, TensorFlow, Fast API, React, PSNR, SSIM, Adapter Head, Residual Learning, SOTS Dataset, Tiled Inference, Deep Learning

INTRODUCTION

Atmospheric phenomena such as haze, fog, and smog significantly degrade the quality of outdoor images by scattering light through particles like dust and moisture. This leads to reduced visibility, poor contrast, and color distortion, which negatively impacts various computer vision applications such as autonomous driving, surveillance systems, and remote sensing.

Traditional image dehazing techniques like Dark Channel Prior (DCP) work under certain assumptions and fail in complex scenarios such as dense or non-uniform haze. With the advancement of deep learning, more robust and accurate methods have emerged.

This project presents an AI-based solution that uses deep learning to remove haze from images effectively. It combines a pre-trained model with a lightweight adaptive component and deploys it as a complete web application for real-world usability.

MOTIVATION

The need for efficient image dehazing arises from real-world problems, especially in pollution-prone regions where visual data is often degraded. Poor image quality reduces the accuracy of systems used in traffic monitoring, agriculture, and security.

Existing solutions are either:

- Cloud-based (costly and dependent on internet), or
- Complex academic models (difficult to use)

This project aims to:

- Provide a **cost-effective local solution**
- Ensure **data privacy (no cloud dependency)**
- Deliver a **user-friendly interface**
- Work efficiently even with limited computational resources

OBJECTIVES

The main objectives of the project are:

- Develop an advanced image dehazing model using **MAXIM-S2 backbone**
- Integrate a **lightweight adapter head** for efficient fine-tuning
- Enable processing of **high-resolution images**
- Build a **full-stack application** (React + FastAPI)
- Compute image quality metrics like **PSNR, SSIM, and MSE**

- Ensure scalability, usability, and extensibility
- Support **offline/local deployment**

These objectives ensure both technical performance and practical usability.

METHODOLOGY

1. System Architecture

The system follows a **three-tier architecture**:

1. **Frontend (React):**
 - User uploads images via drag-and-drop
 - Displays original vs dehazed images
 - Shows performance metrics
2. **Backend (Fast API):**
 - Handles API requests
 - Processes images
 - Computes metrics
3. **Model Layer (TensorFlow):**
 - Performs deep learning inference
 - Uses pre-trained MAXIM-S2 model

2. Model Architecture

MAXIM-S2 Backbone

- Multi-scale encoder-decoder architecture
- Uses MLP-based feature mixing
- Captures both local and global features
- Pre-trained on SOTS dataset
- Parameters are frozen during training

Adapter Head

- Lightweight network (~11,000 parameters)
- Learns residual corrections
- Initialized with zero weights for stability
- Final output:

Final Output = Backbone Output + Residual Correction

3. Tiled Inference Engine

One of the key innovations of this project is the **tiled inference technique**:

- Large images are divided into smaller **256×256 tiles**
- Each tile is processed independently
- Tiles are recombined to form final output

Advantages:

- Supports high-resolution images (even 4K)
- Reduces memory usage
- Maintains image quality

4.4 Training Methodology

- Dataset: **SOTS outdoor dataset (413 image pairs)**
- Preprocessing:
 - Resize images to 256×256
 - Normalize pixel values
- Loss Function:
L1 Loss + 0.2 × (1 – SSIM)
- Optimizer: Adam
- Epochs: 5–10
- Batch Size: 8

EXPERIMENTATION AND RESULTS

1. Performance Metrics

Metric	Hazy vs Ground Truth (Baseline)	Dehazed vs Ground Truth (Our Output)	Improvement
PSNR (dB)	15–20 dB	24–30 dB	+8–12 dB
SSIM	0.60–0.75	0.88–0.95	+0.15–0.20
MSE	250–500	50–120	~5× reduction

2. Observations

- Significant improvement in clarity and visibility
- Better edge sharpness and colour restoration
- Reduced noise and distortion
- Improved structural similarity

3. Performance Efficiency

- CPU inference: ~0.5–1.0 seconds
- GPU inference: ~0.2–0.3 seconds
- Fast response due to model caching
- Efficient handling of large images

CHALLENGES AND SOLUTIONS

Challenge	Solution
Model trained only on 256×256	Tiled inference engine
Slow model loading (~2–3s)	Load once at startup, cache in memory
CORS blocked in local dev	CORS Middleware with allow_origins=["*"]
Random adapter init causes instability	Zero-weight initialization
GPU memory limits for large images	Tile-by-tile processing
Diverse input formats	Standardized RGB preprocessing pipeline

CONCLUSION

The project successfully demonstrates a high-quality AI-based image dehazing system that is:

- Efficient and lightweight
- Cost-effective (no cloud required)
- Scalable and user-friendly
- Suitable for real-world applications

By combining a powerful pre-trained model with a lightweight adapter, the system achieves strong performance even with limited data.

FUTURE WORK

- Video dehazing (real-time processing)
- Mobile application development
- Docker-based deployment
- Multi-dataset training (fog, smoke, indoor haze)
- Model compression for edge devices

Full Signature of Supervisor/s:

.....

Full signature of the student:

.....